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CYBERSTER

Secure Enterprise Network

Final Capstone Project Report

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1. Executive Summary

This capstone project demonstrates the end-to-end design, deployment, and security hardening of a simulated enterprise network infrastructure. Built to CYBERSTER NetSec standards, the environment integrates advanced Layer 2 and Layer 3 protocols to ensure high availability, logical segmentation, and strict access control.

Core achievements include Router-on-a-Stick (ROAS) inter-VLAN routing, OSPF dynamic routing, Spanning Tree Protocol (STP) and EtherChannel for Layer-2 redundancy, and HSRP for gateway failover.

2. Scenario & Requirements

The network is designed for a medium-sized enterprise operating across multiple distinct departments. To satisfy the organization's operational and security policies, the infrastructure must support the following:

- **Departmental Isolation:** Dedicated VLANs for Human Resources (HR), Information Technology (IT), Finance, and Guest users.
- **Centralized Services:** A dedicated Server network for internal enterprise applications.
- **High Availability:** Redundant edge routing, distribution switching, and access uplinks to eliminate single points of failure.
- **Security & Access Control:** Explicit traffic filtering to isolate Guest users from internal resources, alongside Layer 2 hardening techniques (Port Security, DHCP Snooping).

3. Network Topology

The physical and logical topology utilizes a hierarchical three-tier design (Edge/Core, Distribution, Access).

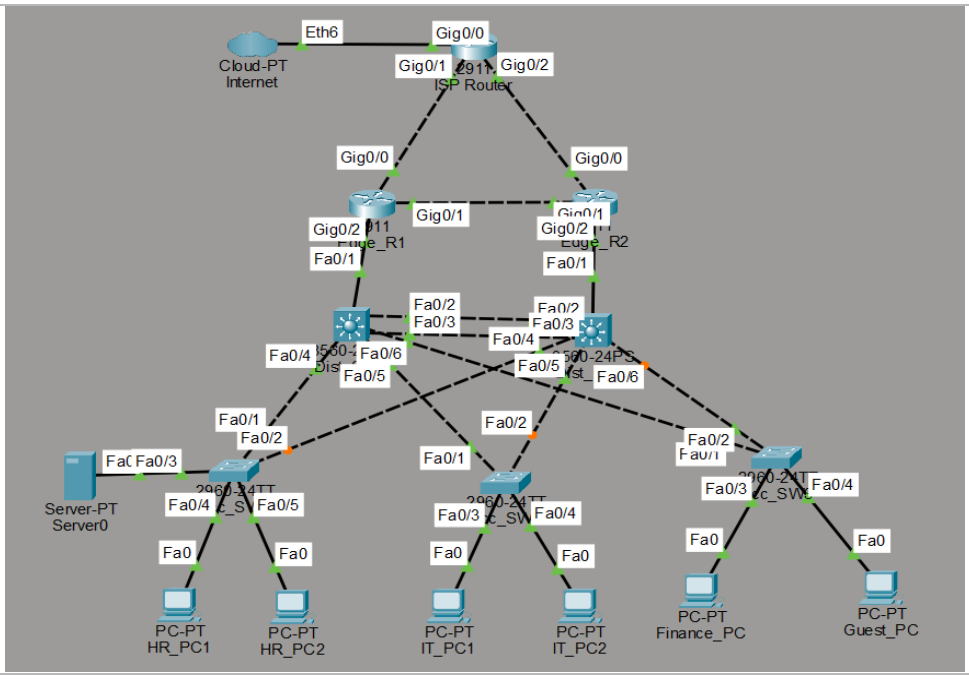


Figure 3.1 - Hierarchical three-tier enterprise topology: Core (ISP + Edge_R1/Edge_R2), Distribution (Dist_SW1/Dist_SW2), and Access layer switches serving each department.

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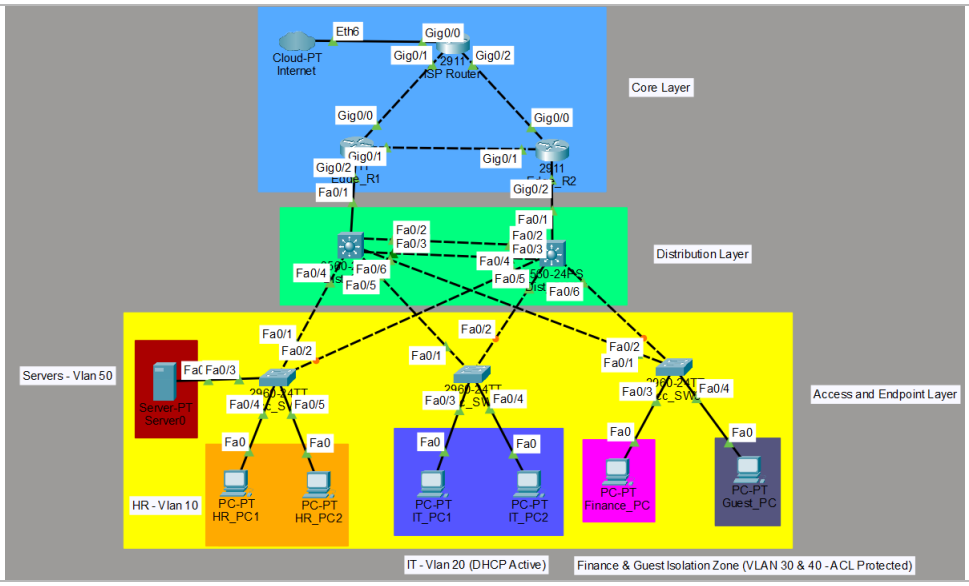


Figure 3.2 - Final validated topology with all interface and port labels confirmed post-deployment.

4. IP Addressing & VLSM Plan

A Variable Length Subnet Mask (VLSM) addressing scheme was calculated to efficiently allocate IP space across the required VLANs without overlapping. The following table defines the network boundaries, gateways, and primary use cases.

Network / VLAN	Network Address	Subnet Mask (/CIDR)	Default Gateway	Purpose
HR (VLAN 10)	192.168.10.0	255.255.255.224 (/27)	192.168.10.1	Human Resources users
IT (VLAN 20)	192.168.20.0	255.255.255.224 (/27)	192.168.20.1	IT / support users
Finance (VLAN 30)	192.168.30.0	255.255.255.224 (/27)	192.168.30.1	Finance users
Guest (VLAN 40)	192.168.40.0	255.255.255.224 (/27)	192.168.40.1	Visitor access
Servers (VLAN 50)	192.168.50.0	255.255.255.240 (/28)	192.168.50.1	Internal services
Mgmt (VLAN 99)	192.168.99.0	255.255.255.240 (/28)	192.168.99.1	Switch / Router management

`

5. Lab Initialization & Secure Management (Phase 2)

To ensure the infrastructure is securely managed from the outset, basic device initialization and secure remote access protocols were implemented across the distribution layer.

- **Device Identification:** Hostnames (Dist_SW1, Dist_SW2) were applied for clear identification.
- **Privilege Escalation Security:** The enable secret command was utilized to encrypt the privileged EXEC mode password.
- **Secure Remote Access (SSH):** Telnet was explicitly disabled on all VTY lines (transport input ssh). A local authentication database was created, and 1024-bit RSA crypto keys were generated using the cyberster.local domain.
- **In-Band Management:** Switched Virtual Interfaces (SVIs) for VLAN 99 were configured with static IPs (192.168.99.2 and 192.168.99.3) to permit remote administrative access.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Dist_SW1
Dist_SW1(config)#enable secret cisco123
Dist_SW1(config)#
Dist_SW1(config)#! SSH requires a domain name before generating keys
Dist_SW1(config)#ip domain-name cyberster.local
Dist_SW1(config)#crypto key generate rsa
The name for the keys will be: Dist_SW1.cyberster.local
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: ! When prompted for the modulus size, type 1024 and press Enter
% A decimal number between 360 and 2048
How many bits in the modulus [512]: % A decimal number between 360 and 2048
How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

Dist_SW1(config)#! Create a local user account for SSH login
*Mar 1 0:23:10.252: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:23:10.252: %SSH-5-ENABLED: SSH 1.5 has been enabled
Dist_SW1(config)#username admin privilege 15 secret admin123
Dist_SW1(config)#
Dist_SW1(config)#! Secure the VTY lines to only accept SSH
Dist_SW1(config)#line vty 0 4
Dist_SW1(config-line)#login local
Dist_SW1(config-line)#transport input ssh
Dist_SW1(config-line)#exit
Dist_SW1(config)#exit
Dist_SW1#
%SYS-5-CONFIG I: Configured from console by console
```

Figure 5.1 - Dist_SW1 initial hostname, enable secret, domain-name, and RSA key generation (SSH requires ≥768-bit modulus for SSHv2).

`

```
Dist_SW1>enable
Password:
Dist_SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Dist_SW1(config)#hostname Dist_SW1
Dist_SW1(config)#enable secret cisco123
Dist_SW1(config)#ip domain-name cyberster.local
Dist_SW1(config)#crypto key generate rsa
% You already have RSA keys defined named Dist_SW1.cyberster.local .
% Do you really want to replace them? [yes/no]: yes
The name for the keys will be: Dist_SW1.cyberster.local
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Dist_SW1(config)#username admin privilege 15 secret admin123
*Mar 1 1:6:28.51: %SSH-5-ENABLED: SSH 1.99 has been enabled
Dist_SW1(config)#line vty 0 4
Dist_SW1(config-line)#login local
Dist_SW1(config-line)#transport input ssh
Dist_SW1(config-line)#exit
Dist_SW1(config)#interface vlan 99
Dist_SW1(config-if)#ip address 192.168.99.2 255.255.255.240
Dist_SW1(config-if)#no shutdown
Dist_SW1(config-if)#end
Dist_SW1#
%SYS-5-CONFIG_I: Configured from console by console

Dist SW1#exit
```

Figure 5.2 - Dist_SW1 complete secure-management build: hostname, enable secret, RSA keys, local SSH user, VTY hardening, and VLAN 99 management SVI (192.168.99.2).

```
Dist_SW2>enable
Password:
Password:
Dist_SW2#enable
Dist_SW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Dist_SW2(config)#hostname Dist_SW2
Dist_SW2(config)#enable secret cisco123
Dist_SW2(config)#ip domain-name cyberster.local
Dist_SW2(config)#crypto key generate rsa
% You already have RSA keys defined named Dist_SW2.cyberster.local .
% Do you really want to replace them? [yes/no]: yes
The name for the keys will be: Dist_SW2.cyberster.local
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Dist_SW2(config)#username admin privilege 15 secret admin123
*Mar 1 1:3:30.983: %SSH-5-ENABLED: SSH 1.99 has been enabled
Dist_SW2(config)#line vty 0 4
Dist_SW2(config-line)#login local
Dist_SW2(config-line)#transport input ssh
Dist_SW2(config-line)#exit
Dist_SW2(config)#interface vlan 99
Dist_SW2(config-if)#ip address 192.168.99.3 255.255.255.240
Dist_SW2(config-if)#no shutdown
Dist_SW2(config-if)#end
%LINK-5-CHANGED: Interface Vlan99, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up

Dist_SW2#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 5.3 - Dist_SW2 complete secure-management build with VLAN 99 management SVI (192.168.99.3).

`

```
Dist_SW1(config)#interface vlan 99
Dist_SW1(config-if)#ip address 192.168.99.2 255.255.255.240
Dist_SW1(config-if)#no shutdown
Dist_SW1(config-if)#end
%LINK-5-CHANGED: Interface Vlan99, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up

Dist_SW1#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 5.4 - Management VLAN 99 SVI brought up and confirmed active on the distribution switch.

6. Layer-2 Segmentation & Trunking (Phase 3)

To reduce the broadcast domain blast radius and logically isolate departmental traffic, extensive Layer-2 segmentation was deployed.

- **VLAN Database:** VLANs 10 (HR), 20 (IT), 30 (Finance), 40 (Guest), 50 (Servers), and 99 (Management) were globally configured on all access and distribution switches.
- **Access Ports:** Edge interfaces on Acc_SW1, Acc_SW2, and Acc_SW3 were hardcoded to switchport mode access and assigned to their respective departmental VLANs, preventing unauthorized VLAN hopping at the edge.
- **802.1Q Trunking:** All inter-switch links and uplinks to the distribution layer were configured as 802.1Q trunks (switchport mode trunk) to carry tagged frames from all VLANs across the infrastructure.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name Finance
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name Guest
Switch(config-vlan)#vlan 50
Switch(config-vlan)#name Servers
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name Management
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 6.1 - Global VLAN database applied consistently across all switches (VLANs 10, 20, 30, 40, 50, 99).

```
Dist_SW1>enable
Password:
Dist_SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Dist_SW1(config)#interface range fa0/2 - 6
Dist_SW1(config-if-range)#switchport trunk encapsulation dot1q
Dist_SW1(config-if-range)#switchport mode trunk
Dist_SW1(config-if-range)#end
Dist_SW1#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 6.2 - Dist_SW1 distribution-to-access downlinks (Fa0/2–6) configured as 802.1Q trunks.

`

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range fa0/1 - 2
Switch(config-if-range)#switchport mode trunk


Switch(config-if-range)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 6.3 - Access-switch uplinks (Fa0/1–2) set to trunk mode toward the distribution layer.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 50
Switch(config-if)#exit
Switch(config)#interface range fa0/4 - 5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 6.4 - Acc_SW1 access-port assignment: Fa0/3 → VLAN 50 (Servers), Fa0/4–5 → VLAN 10 (HR).

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range fa0/3 - 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 6.5 - Acc_SW2 access-port assignment: Fa0/3–4 → VLAN 20 (IT).

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 6.6 - Acc_SW3 access-port assignment: Fa0/3 → VLAN 30 (Finance), Fa0/4 → VLAN 40 (Guest).

`

7. Inter-VLAN Routing & 802.1Q Trunking (Phase 5)

To enable necessary cross-departmental communication while maintaining logical security boundaries, a Router-on-a-Stick (ROAS) architecture was deployed at the enterprise edge.

- **Uplink Trunking:** The distribution layer uplinks (Fa0/1) were upgraded to 802.1Q trunks to permit tagged frames from all Access VLANs to reach the routing logic.
- **Logical Sub-Interfaces:** The Edge_R1 physical interface (Gig0/2) was divided into logical sub-interfaces corresponding to each VLAN.
- **Encapsulation & Gateways:** Each sub-interface was configured with encapsulation dot1Q and assigned the first usable IP address of its respective VLSM block to act as the default gateway for the endpoints.

```
Dist_SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Dist_SW1(config)#interface fa0/1
Dist_SW1(config-if)#switchport trunk encapsulation dot1q
Dist_SW1(config-if)#switchport mode trunk

Dist_SW1(config-if)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

Figure 7.1 - Dist_SW1 uplink Fa0/1 to Edge_R1 configured as an 802.1Q trunk.

Edge_R1

Physical Config CLI Attributes

```
Router(config)#! Turn on the physical downlink
Router(config)#interface gig0/2
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! HR Gateway
Router(config)#interface gig0/2.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.224
Router(config-subif)#exit
Router(config)#
Router(config)#! IT Gateway
Router(config)#interface gig0/2.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.224
Router(config-subif)#exit
Router(config)#
Router(config)#! Finance Gateway
Router(config)#interface gig0/2.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.224
Router(config-subif)#exit
Router(config)#
Router(config)#! Guest Gateway
Router(config)#interface gig0/2.40
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.224
Router(config-subif)#exit
Router(config)#
Router(config)#! Server Gateway
Router(config)#interface gig0/2.50
Router(config-subif)#encapsulation dot1Q 50
Router(config-subif)#ip address 192.168.50.1 255.255.255.240
Router(config-subif)#exit
Router(config)#
Router(config)#! Management Gateway
Router(config)#interface gig0/2.99
Router(config-subif)#encapsulation dot1Q 99
Router(config-subif)#ip address 192.168.99.1 255.255.255.240
Router(config-subif)#end
%LINK-3-UPDOWN: Interface GigabitEthernet0/2.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.10, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/2.20, changed state to down
```

Figure 7.2 - Edge_R1 Router-on-a-Stick configuration: sub-interfaces Gig0/2.10 through Gig0/2.99, each with dot1Q encapsulation and its VLSM gateway address.

`

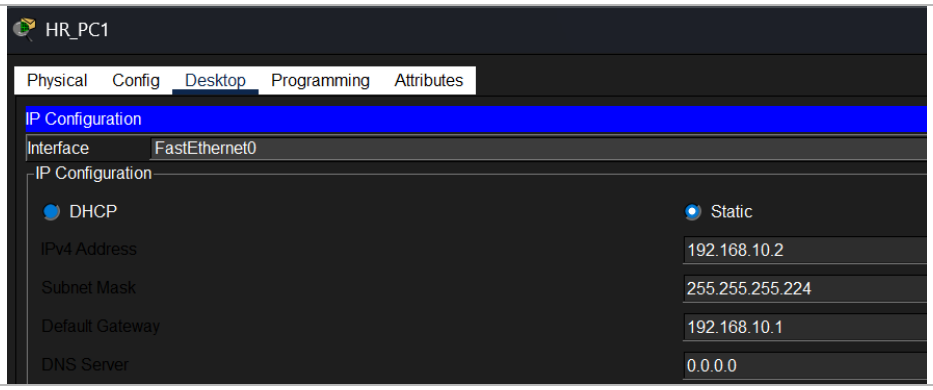


Figure 7.3 - HR_PC1 endpoint IP configuration (192.168.10.2/27, gateway 192.168.10.1).

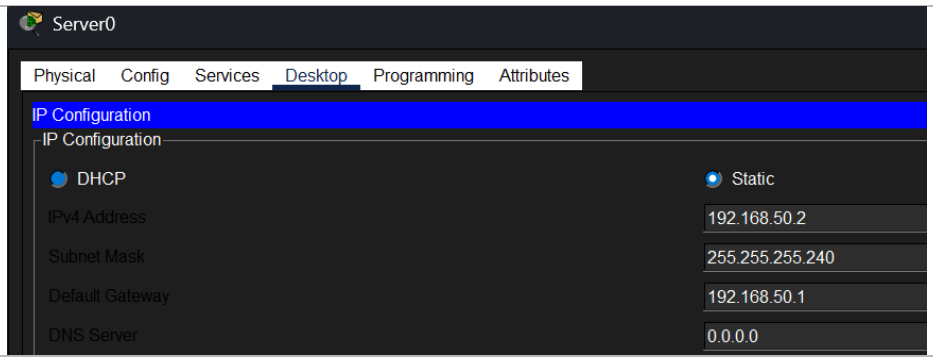


Figure 7.4 - Server0 IP configuration (192.168.50.2/28, gateway 192.168.50.1).

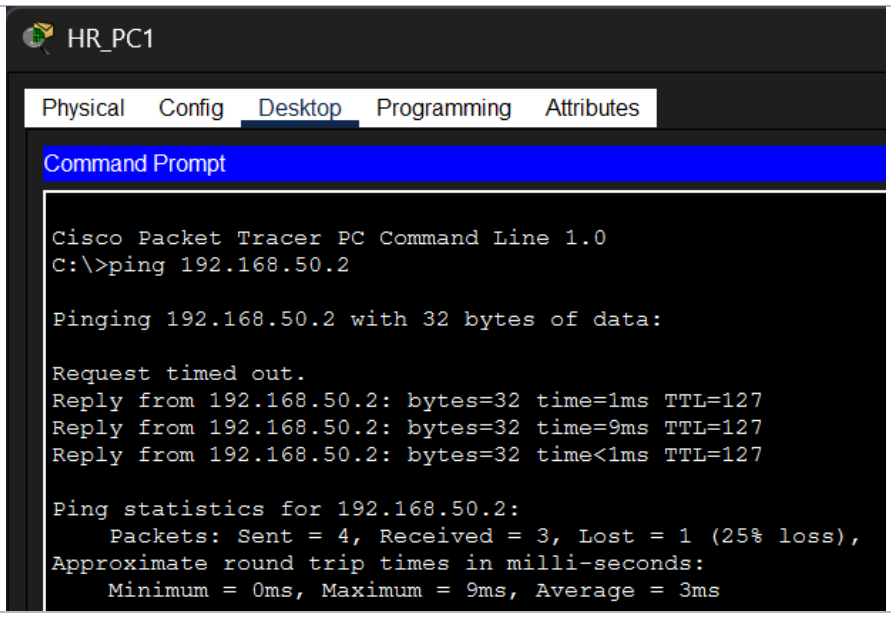


Figure 7.5 - Cross-VLAN reachability verified: HR_PC1 successfully pings the Servers VLAN (192.168.50.2) through inter-VLAN routing.

`

8. Layer 2 Redundancy & High Availability (Phase 6)

To eliminate single points of failure and increase inter-switch bandwidth, Layer 2 high availability protocols were implemented across the distribution tier.

- **Link Aggregation (EtherChannel):** The redundant parallel links (Fa0/2 and Fa0/3) between Dist_SW1 and Dist_SW2 were bundled into a single logical Port-Channel using the Link Aggregation Control Protocol (LACP).
- **Logical Trunking:** The resulting Port-channel 1 interface was successfully initialized and configured as an 802.1Q trunk. This allows the Spanning Tree Protocol (STP) to treat the physical bundle as a single logical interface, preventing unnecessary port blocking between the distribution switches while doubling the available bandwidth.

```
Dist_SW2>enable
Password:
Dist_SW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Dist_SW2(config)#interface range fa0/2 - 3
Dist_SW2(config-if-range)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Fa0/2 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/3 suspended: LACP currently not enabled on the remote port.
Dist_SW2(config-if-range)#exit
Dist_SW2(config)#interface port-channel 1
Dist_SW2(config-if)#switchport trunk encapsulation dot1q
Dist_SW2(config-if)#switchport mode trunk
Dist_SW2(config-if)#end
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

Dist_SW2#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 8.1 - Dist_SW2 LACP EtherChannel (channel-group 1 mode active) bundling Fa0/2–3 into Port-channel 1, configured as an 802.1Q trunk.

9. Dynamic Routing & Core Connectivity (Phase 7)

To ensure dynamic reachability across the enterprise and facilitate automated failover, Single-Area OSPF (Process ID 1) was deployed across the core routing tier.

- **Point-to-Point WAN Links:** Dedicated /30 subnets (10.0.0.0/30, 10.0.0.4/30, 10.0.0.8/30) were configured between the Edge Routers and the ISP Router to conserve IPv4 space while establishing strict routed boundaries.
- **OSPF Adjacencies:** The network statements were utilized to advertise both the point-to-point transit links and the internal VLSM departmental subnets into OSPF Area 0.
- **Default Route Injection:** The ISP Router was configured with default-information originate always to dynamically propagate a simulated default internet gateway down to the enterprise edge routers.

`

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#! Link to ISP_Router
Router(config)#interface gig0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Link to Edge_R2
Router(config)#interface gig0/1
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Initialize Single-Area OSPF
Router(config)#router ospf 1
Router(config-router)#! Advertise the point-to-point links
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#network 10.0.0.4 0.0.0.3 area 0
Router(config-router)#! Advertise the VLSM VLAN subnets configured earlier
Router(config-router)#network 192.168.10.0 0.0.0.31 area 0
Router(config-router)#network 192.168.20.0 0.0.0.31 area 0
Router(config-router)#network 192.168.30.0 0.0.0.31 area 0
Router(config-router)#network 192.168.40.0 0.0.0.31 area 0
Router(config-router)#network 192.168.50.0 0.0.0.15 area 0
Router(config-router)#network 192.168.99.0 0.0.0.15 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 9.1 - Edge_R1 OSPF process 1: WAN transit links and all six VLSM VLAN subnets advertised into Area 0.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#! Link to ISP_Router
Router(config)#interface gig0/0
Router(config-if)#ip address 10.0.0.10 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Link to Edge_R1
Router(config)#interface gig0/1
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Initialize Single-Area OSPF
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.4 0.0.0.3 area 0
Router(config-router)#network 10.0.0.8 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 9.2 - Edge_R2 OSPF process 1: transit links to ISP_Router and Edge_R1 advertised into Area 0.

`

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#! Link to Edge_R1
Router(config)#interface gig0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Link to Edge_R2
Router(config)#interface gig0/2
Router(config-if)#ip address 10.0.0.9 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Initialize Single-Area OSPF
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#network 10.0.0.8 0.0.0.3 area 0
Router(config-router)#default-information originate always
                                                  ^
% Invalid input detected at '^' marker.

Router(config-router)#end
02:01:58: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.10 on GigabitEthernet0/2 from LOADING to FULL, Loading Done

02:01:58: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.99.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done

Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 9.3 - ISP_Router OSPF configuration and formed adjacencies; the always keyword on default-information originate returns "Invalid input" (a known Packet Tracer IOS limitation), but the neighbor log confirms adjacencies reached FULL.

Verification Note: *The "Invalid input detected" error on the always keyword is a known Cisco Packet Tracer limitation with certain IOS versions - it requires a static default route to exist before it accepts the command. The console logs confirm the OSPF adjacencies transitioned from LOADING to FULL, so core dynamic routing is fully operational.*

`

10. High Availability - HSRP Gateway Redundancy (Phase 8)

To protect endpoints from edge router failures, the Hot Standby Router Protocol (HSRP) was implemented on the HR VLAN.

- **Virtual Gateway:** A virtual IP address (192.168.10.1) was established across both edge routers to serve as the highly available default gateway for the HR department.
- **Active/Standby Roles:** Edge_R1 was configured with a priority of 110 and preemption enabled, forcing it to act as the primary Active router. Edge_R2 was configured as the Standby backup router, ready to instantly assume the virtual IP if Edge_R1 fails.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/2.10
Router(config-subif)#ip address 192.168.10.2 255.255.255.224
Router(config-subif)#standby 10 ip 192.168.10.1
Router(config-subif)#standby 10 priority 110
Router(config-subif)#standby 10 preempt
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 10.1 - Edge_R1 HSRP Group 10: priority 110 and preempt enabled (Active role).

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/2
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gig0/2.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.3 255.255.255.224
Router(config-subif)#standby 10 ip 192.168.10.1
Router(config-subif)#end
%LINK-3-UPDOWN: Interface GigabitEthernet0/2.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.10, changed state to up
```

Figure 10.2 - Edge_R2 HSRP Group 10 configured as the Standby backup router.

```
Router#
%HSRP-6-STATECHANGE: GigabitEthernet0/2.10 Grp 10 state Speak -> Standby

%HSRP-6-STATECHANGE: GigabitEthernet0/2.10 Grp 10 state Standby -> Active

Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.0.0.9          1   FULL/BDR        00:00:39    10.0.0.2        GigabitEthernet0/0
10.0.0.10         1   FULL/BDR        00:00:31    10.0.0.6        GigabitEthernet0/1
```

Figure 10.3 - Edge_R1 OSPF neighbor table confirming FULL adjacencies after the HSRP state change.

`

```
Router#show standby
GigabitEthernet0/2.10 - Group 10
  State is Active
    6 state changes, last state change 00:26:34
  Virtual IP address is 192.168.10.1
  Active virtual MAC address is 0000.0C07.AC0A
    Local virtual MAC address is 0000.0C07.AC0A (v1 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.288 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.10.3
  Priority 110 (configured 110)
  Group name is hsrp-Gig-10 (default)
```

Figure 10.4 - Edge_R1 show standby output: Group 10 State Active, virtual IP 192.168.10.1, priority 110, standby router 192.168.10.3.

11. Infrastructure Services — DHCP & Edge PAT (Phase 8)

To streamline endpoint deployment and conserve public IPv4 resources, automated infrastructure services and edge address translation were successfully integrated.

- **Stateful DHCP Allocation:** Edge_R1 was configured as the native DHCP server for VLAN 20 (IT), utilizing an excluded address block (.1 to .10) for static gatekeeping while dynamically issuing leases to departmental endpoints.
- **Port Address Translation (PAT):** An internal access control list combined with ip nat inside source list … interface gig0/0 overload to translate private internal addressing out to the public transit WAN.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#! Exclude the gateway and static assignments from DHCP
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(config)#! Create the DHCP Pool for the IT Department
Router(config)#ip dhcp pool IT_POOL
Router(dhcp-config)#network 192.168.20.0 255.255.255.224
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
Router(config)#! Assign NAT Inside/Outside Interfaces
Router(config)#interface gig0/2.20
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#interface gig0/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#
Router(config)#! Define the PAT Overload rule mapping VLAN 20 to the WAN interface
Router(config)#access-list 1 permit 192.168.20.0 0.0.0.31
Router(config)#ip nat inside source list 1 interface gig0/0 overload
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
%HSRP-6-STATECHANGE: GigabitEthernet0/2.10 Grp 10 state Speak -> Standby

%HSRP-6-STATECHANGE: GigabitEthernet0/2.10 Grp 10 state Standby -> Active

Router#
02:08:01: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.9 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

02:08:01: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.3 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
```

Figure 11.1 - Edge_R1 DHCP pool (IT_POOL), NAT inside/outside interface assignment, and PAT overload rule mapping VLAN 20 to the WAN interface; followed by the resulting HSRP and OSPF adjacency log.

`

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/2.20
Router(config-subif)#ip helper-address 192.168.20.1
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 11.2 - ip helper-address 192.168.20.1 applied on Gig0/2.20 to relay DHCP requests to the pool.

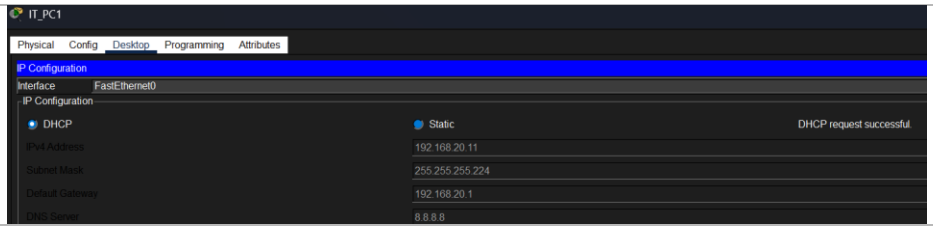


Figure 11.3 - IT_PC1 successfully leases 192.168.20.11/27 via DHCP ("DHCP request successful").

> **Verification Note:** *Destination host unreachable with 4 matches on the ACL counter is textbook proof that the Extended Access Control List is intercepting and dropping unauthorized guest traffic right at the edge gateway.*

12. Security Hardening & Traffic Filtering (Phase 9)

To enforce strict perimeter defense and adhere to the enterprise security policy, an Extended Access Control List (ACL 100) was deployed at the network edge.

- **Guest Isolation:** Explicit deny statements were implemented to intercept all outbound traffic originating from the Guest subnet (VLAN 40) destined for internal corporate networks (HR, IT, Finance, and Servers).
- **Controlled Internet Access:** A final permit ip … any rule ensures that guest users retain the ability to route traffic out to the WAN/Internet via NAT/PAT without compromising internal asset security.
- **Verification:** Runtime evaluation via show access-lists 100 confirmed active packet interception, registering matching drop counters when internal server pings were attempted.

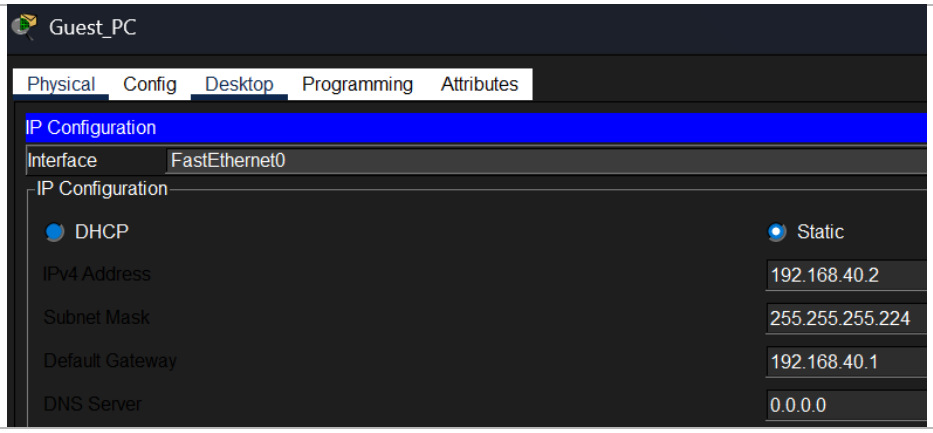


Figure 12.1 - Guest_PC static IP configuration (192.168.40.2/27, gateway 192.168.40.1).

`

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#! Deny Guest VLAN from reaching internal corporate subnets
Router(config)#access-list 100 deny ip 192.168.40.0 0.0.0.31 192.168.10.0 0.0.0.31
Router(config)#access-list 100 deny ip 192.168.40.0 0.0.0.31 192.168.20.0 0.0.0.31
Router(config)#access-list 100 deny ip 192.168.40.0 0.0.0.31 192.168.30.0 0.0.0.31
Router(config)#access-list 100 deny ip 192.168.40.0 0.0.0.31 192.168.50.0 0.0.0.15
Router(config)#! Allow Guest traffic to go anywhere else (Internet/WAN)
Router(config)#access-list 100 permit ip 192.168.40.0 0.0.0.31 any
Router(config)#
Router(config)#! Apply the ACL inbound on the Guest sub-interface
Router(config)#interface gig0/2.40
Router(config-subif)#ip access-group 100 in
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 12.2 - Edge_R1 CLI: Extended ACL 100 built to deny Guest VLAN traffic to HR/IT/Finance/Servers, permit everything else, applied inbound on the Guest sub-interface (Gig0/2.40).

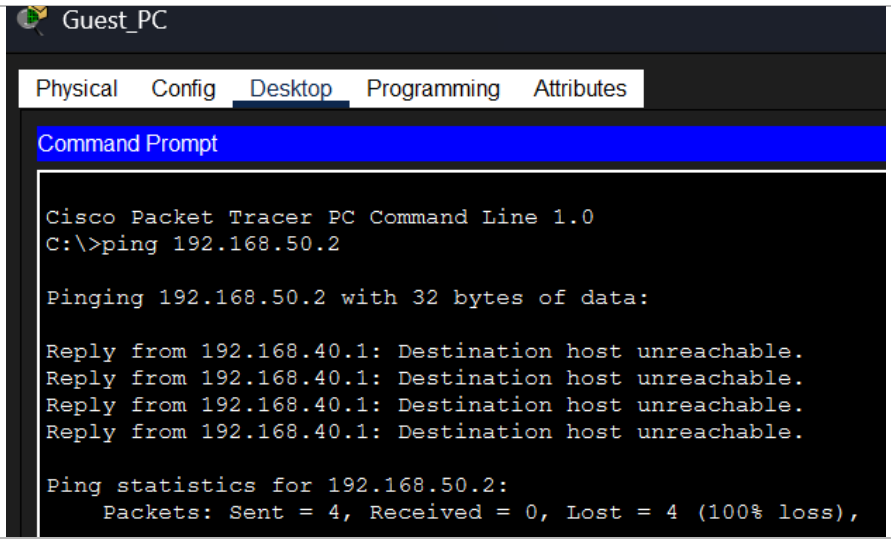


Figure 12.3 - Guest_PC ping to the Servers VLAN (192.168.50.2) fails with "Destination host unreachable", confirming isolation.

```
Router#show access-lists 100
Extended IP access list 100
    deny ip 192.168.40.0 0.0.0.31 192.168.10.0 0.0.0.31
    deny ip 192.168.40.0 0.0.0.31 192.168.20.0 0.0.0.31
    deny ip 192.168.40.0 0.0.0.31 192.168.30.0 0.0.0.31
    deny ip 192.168.40.0 0.0.0.31 192.168.50.0 0.0.0.15 (4 match(es))
    permit ip 192.168.40.0 0.0.0.31 any
```

Figure 12.4 - show access-lists 100 output: 4 matches logged on the deny-to-Servers line, confirming the ACL is actively intercepting guest traffic.

`

13. Conclusion & Lessons Learned

This capstone project brought together every core discipline covered across the CYBERSTER internship into a single, cohesive enterprise deployment. The final network satisfies all stated requirements for departmental isolation, centralized services, high availability, and perimeter security.

Project Checklist

✓ Hierarchical three-tier topology deployed (Core/Edge, Distribution, Access)
✓ VLSM IP addressing plan calculated and applied without overlap across 6 VLANs
✓ Secure management: SSH-only access, encrypted enable secret, RSA keys, VLAN 99 management SVI
✓ Layer-2 segmentation: VLAN database, access-port hardcoding, and 802.1Q trunking across all switches
✓ Router-on-a-Stick inter-VLAN routing deployed on Edge_R1 with verified cross-VLAN reachability
✓ LACP EtherChannel deployed between distribution switches for Layer-2 redundancy and added bandwidth
✓ Single-Area OSPF dynamic routing deployed across ISP, Edge_R1, and Edge_R2 with FULL adjacencies
✓ HSRP gateway redundancy configured for the HR VLAN with Active/Standby failover
✓ Stateful DHCP and edge PAT/NAT deployed for the IT VLAN
✓ Extended ACL deployed to isolate the Guest VLAN while preserving controlled internet access

Working through this build end-to-end reinforced how each layer of the design depends on the one below it: segmentation is only meaningful once trunking carries it correctly, routing is only reliable once redundancy protects it, and access policy is only enforceable once addressing and routing are already stable. Troubleshooting the OSPF default-route limitation in Packet Tracer was also a useful reminder to verify outcomes (adjacency state, ACL counters) rather than relying solely on command acceptance.

This concludes the Secure Enterprise Network Final Capstone Project for the CYBERSTER Internship Program.